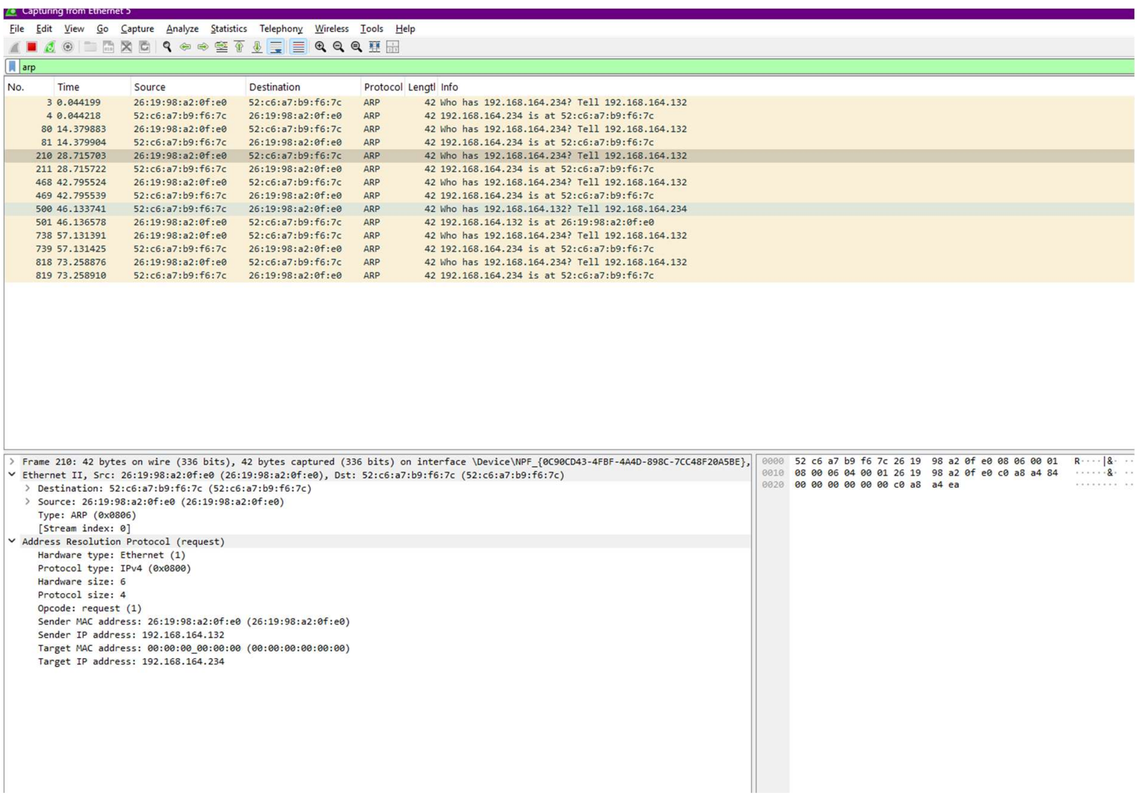
Networking Training Program (Module 6)

1. ARP

- ARP resolves IP addresses to MAC addresses, enabling devices to communicate within a local network.
- It allows routers to determine the next-hop MAC address for packet forwarding.
- ARP caching improves efficiency by reducing repeated address lookups.
- Without ARP, network devices wouldn't know where to send packets, disrupting communication.



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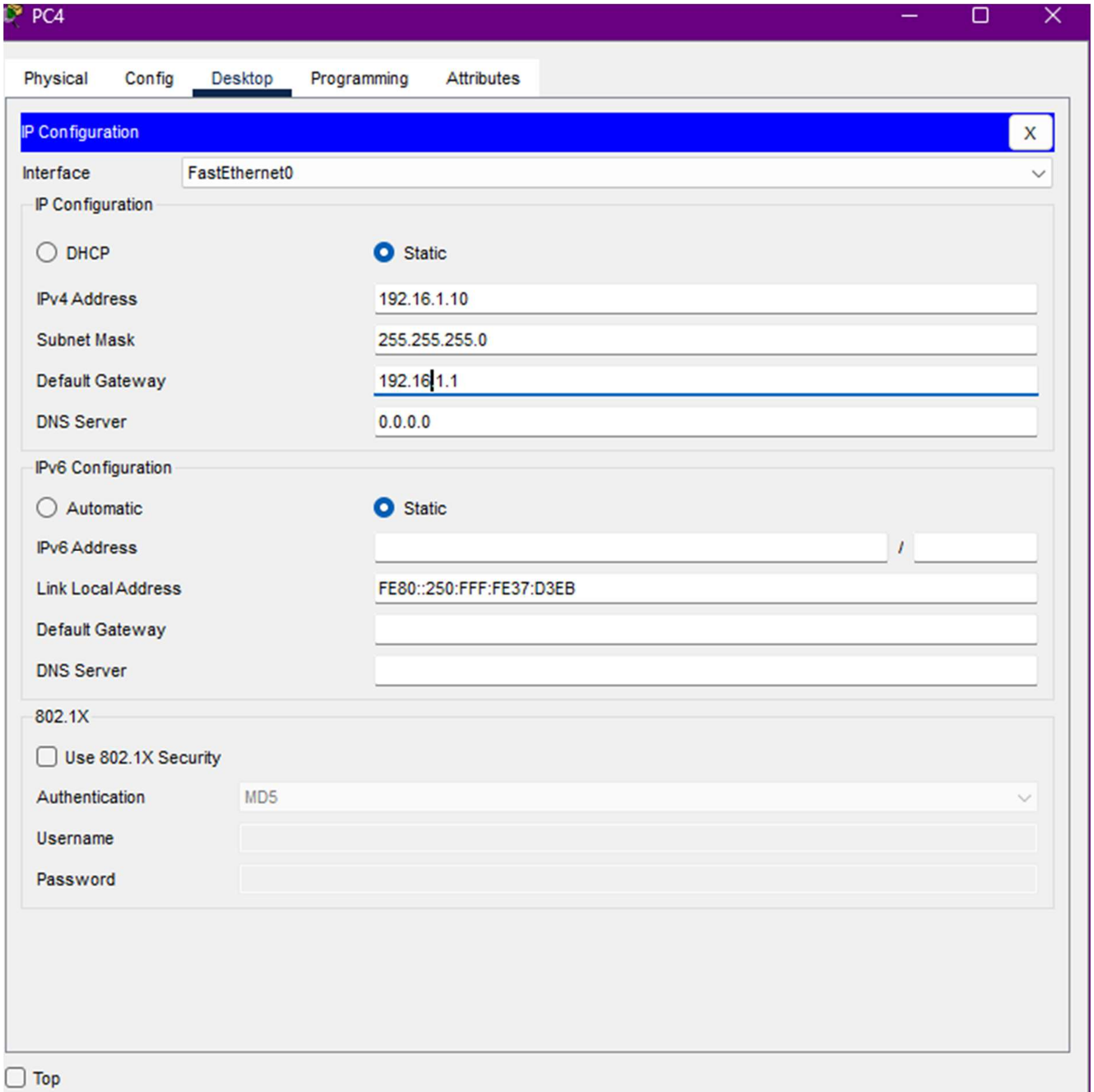
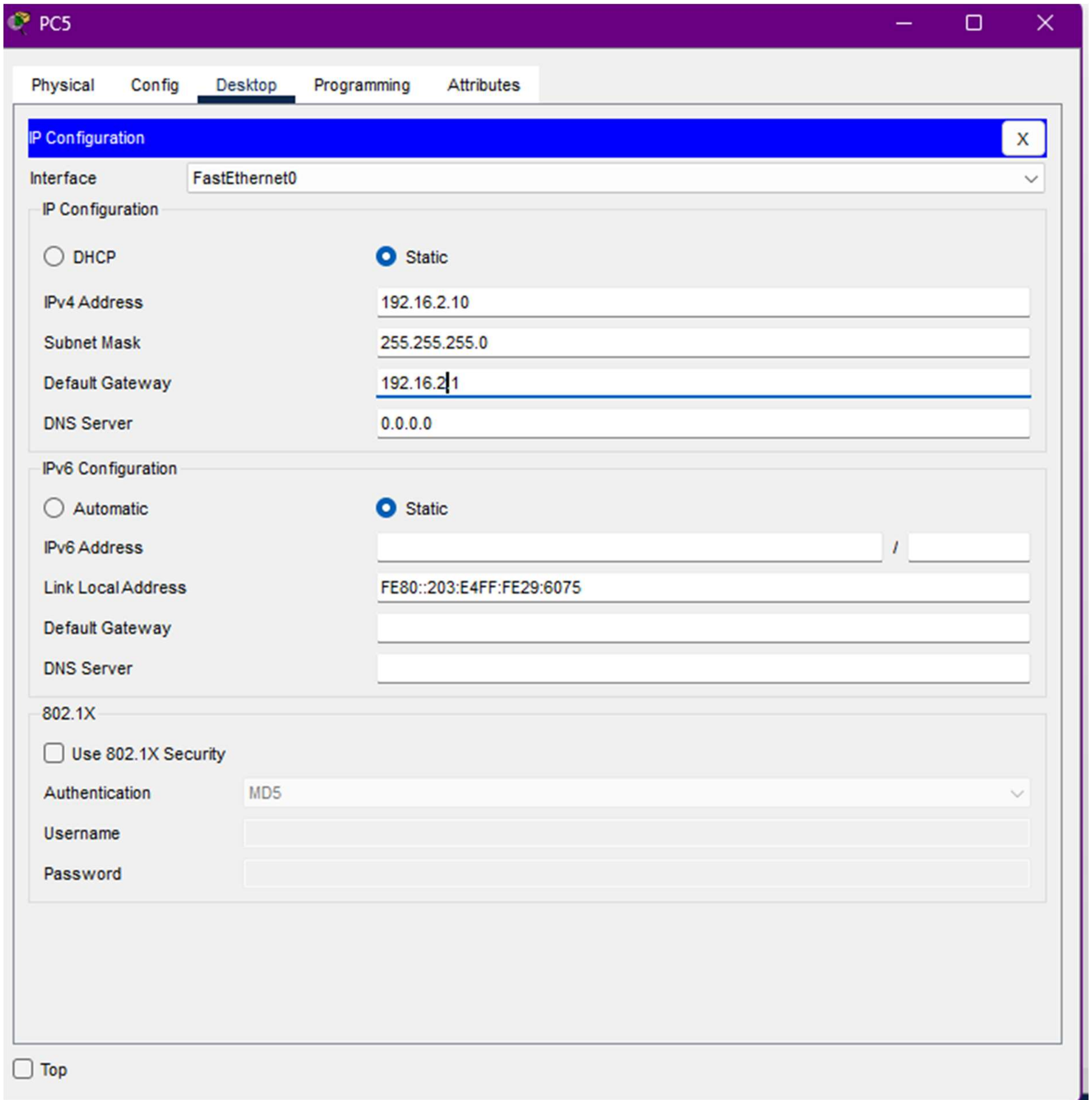
2. Manually configure and Traceroute Tracking


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.16.1.10
Subnet Mask
255.255.255.0
Default Gateway
192.16.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::250:FFF:FE37:D3EB
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.16.2.10
Subnet Mask
255.255.255.0
Default Gateway
192.16.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::203:E4FF:FE29:6075
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

```
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.16.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
%Invalid interface type and number
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.16.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#
```

```
Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#interface gigabitethernet0/0
Router(config-if)#
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

```
Router#
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

```
C:\>traceroute 192.168.2.10
Invalid Command.

C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1
  2   0 ms      0 ms      0 ms      192.168.2.10

Trace complete.
```

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3. Network Subnetting

Subnetting 10.0.0.0/24 into 4 Equal Subnets

1. Given Network Information

- Network Address: 10.0.0.0/24
- Subnet Mask: 255.255.255.0
- Total Hosts in /24: $2^8 - 2 = 254$ hosts
- Requirement: Divide into 4 equal subnets

2. Calculate the New Subnet Mask

- To divide into 4 subnets, we need 2 extra bits ($2^2 = 4$).
- The new subnet mask becomes /26 (255.255.255.192).
- Each subnet has $2^6 - 2 = 62$ usable hosts.

3. Subnet Allocation and Host Ranges

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	10.0.0.0/26	10.0.0.1	10.0.0.62	10.0.0.63
2	10.0.0.64/26	10.0.0.65	10.0.0.126	10.0.0.127
3	10.0.0.128/26	10.0.0.129	10.0.0.190	10.0.0.191
4	10.0.0.192/26	10.0.0.193	10.0.0.254	10.0.0.255

4. Assign IP Addresses to Devices in Packet Tracer

- Router (Default Gateway): Assign first usable IP from each subnet.
- PCs, Switches: Assign IPs within the valid host range.

Example Device IP Assignments

Device	Assigned IP	Subnet Mask	Gateway
Router	10.0.0.1	255.255.255.192	N/A
PC1	10.0.0.2	255.255.255.192	10.0.0.1
PC2	10.0.0.65	255.255.255.192	10.0.0.64
PC3	10.0.0.129	255.255.255.192	10.0.0.128

Device	Assigned IP	Subnet Mask	Gateway
PC4	10.0.0.193	255.255.255.192	10.0.0.192

5. Verify Connectivity

1. Ping Test:
   - Use the ping command to check connectivity between devices in the same and different subnets.
   - Example:

ping 10.0.0.65

2. Traceroute Test:
   - Use tracert or traceroute to check packet paths between different subnets.
   - Example:

tracert 10.0.0.129

3. Check Routing Table (On Router):
4. show ip route

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4. IP Address Classification and NAT Explanation

1. Identifying the Class of Each IP Address

IP addresses are categorized into different classes based on their first octet.

IP Address	First Octet Range	Class
192.168.10.5	192 - 223	Class C
172.20.15.1	128 - 191	Class B
8.8.8.8	1 - 126	Class A

2. Determining if the IP is Private or Public

Some IP ranges are reserved for private networks, while others are public and routable on the internet.

IP Address	Private/Public	Reason
192.168.10.5	Private	Falls in 192.168.0.0 - 192.168.255.255 (Private Class C)
172.20.15.1	Private	Falls in 172.16.0.0 - 172.31.255.255 (Private Class B)
8.8.8.8	Public	Google DNS, routable on the internet

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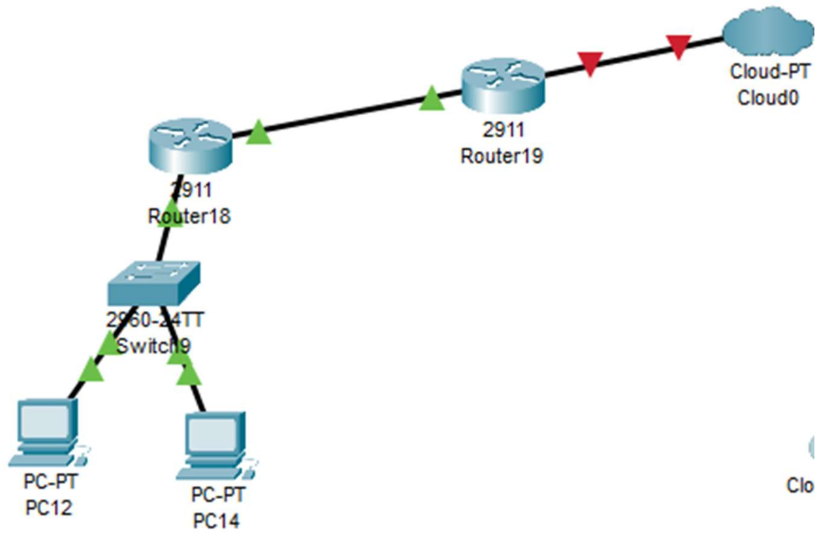
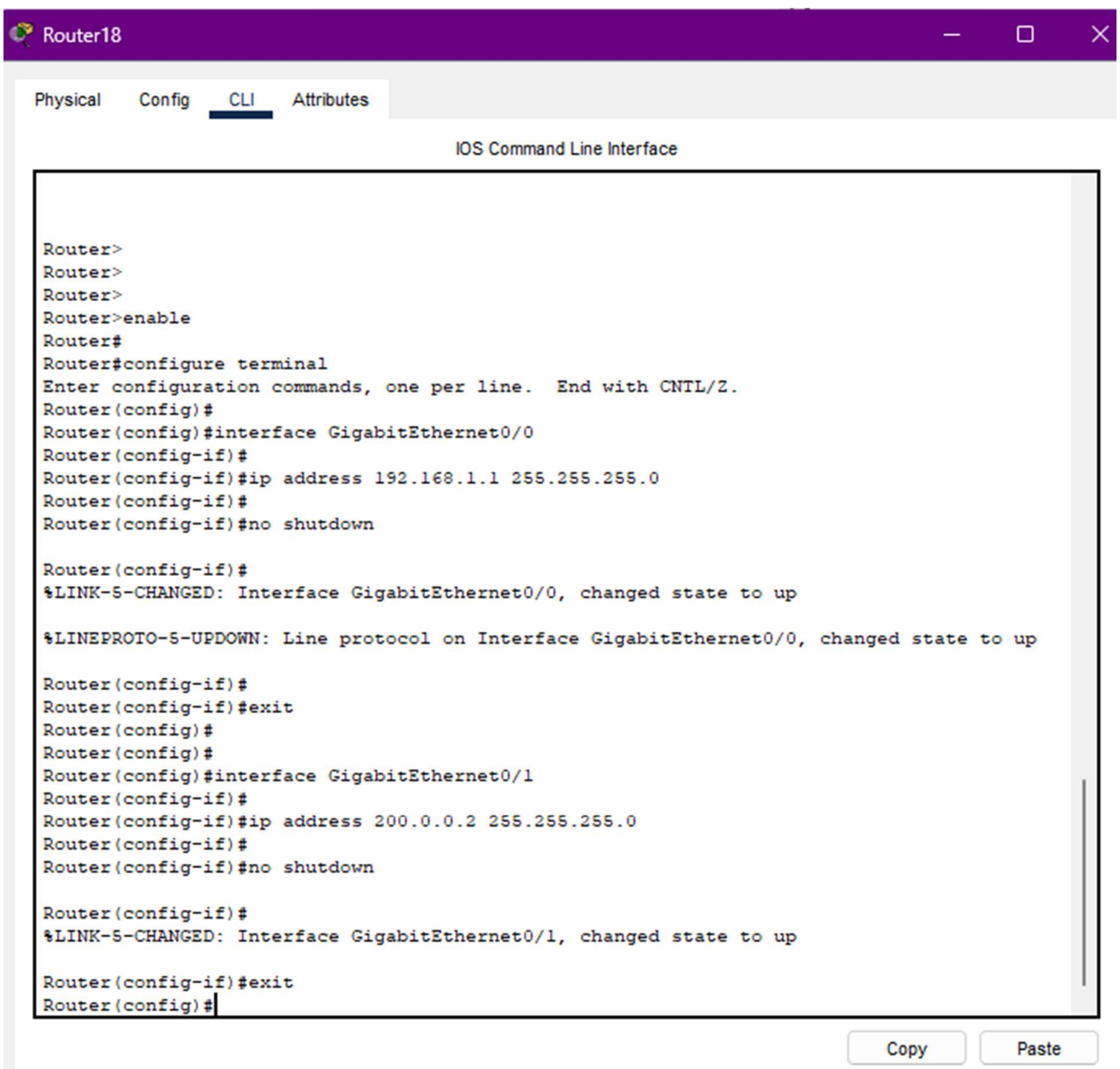
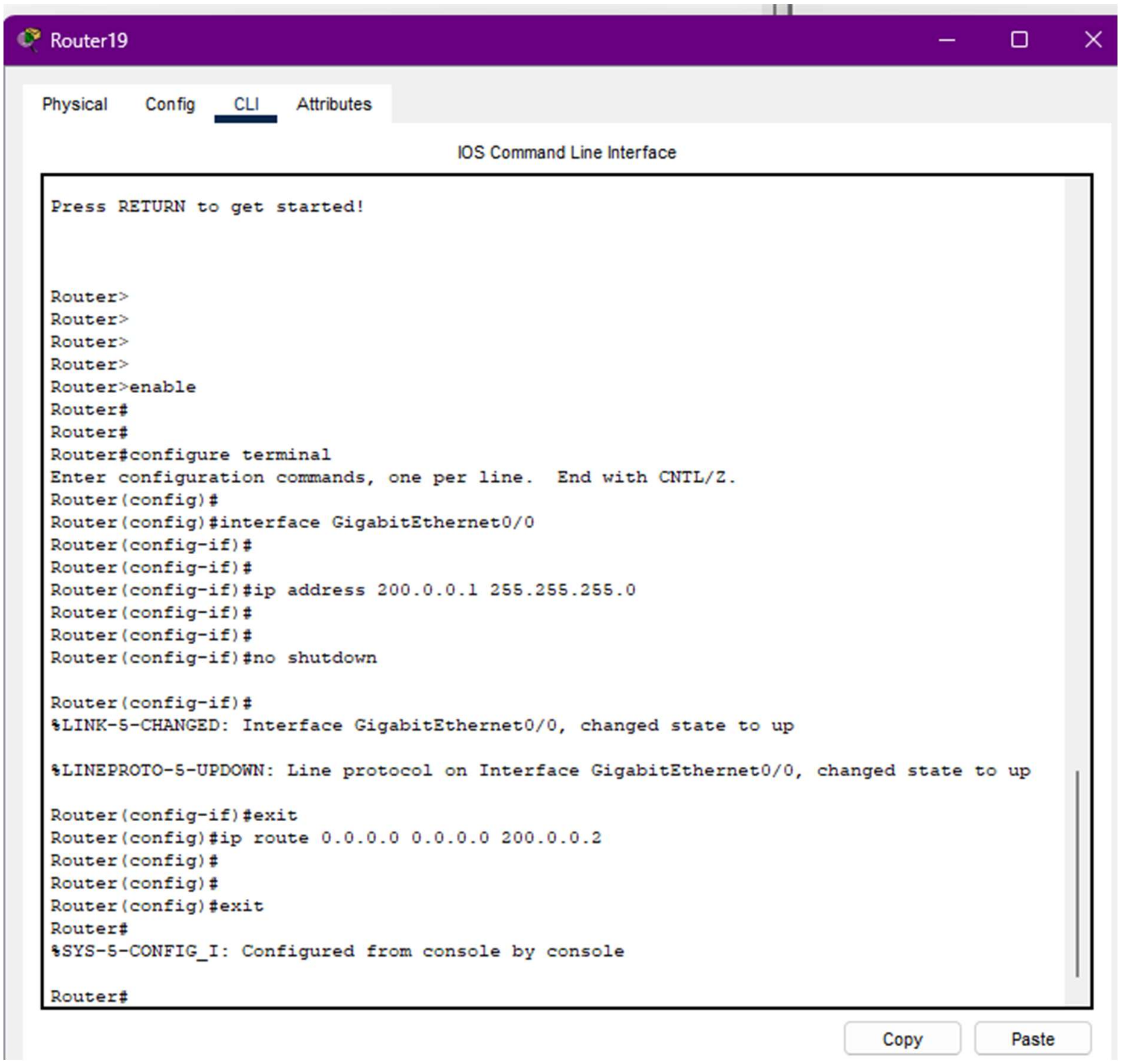
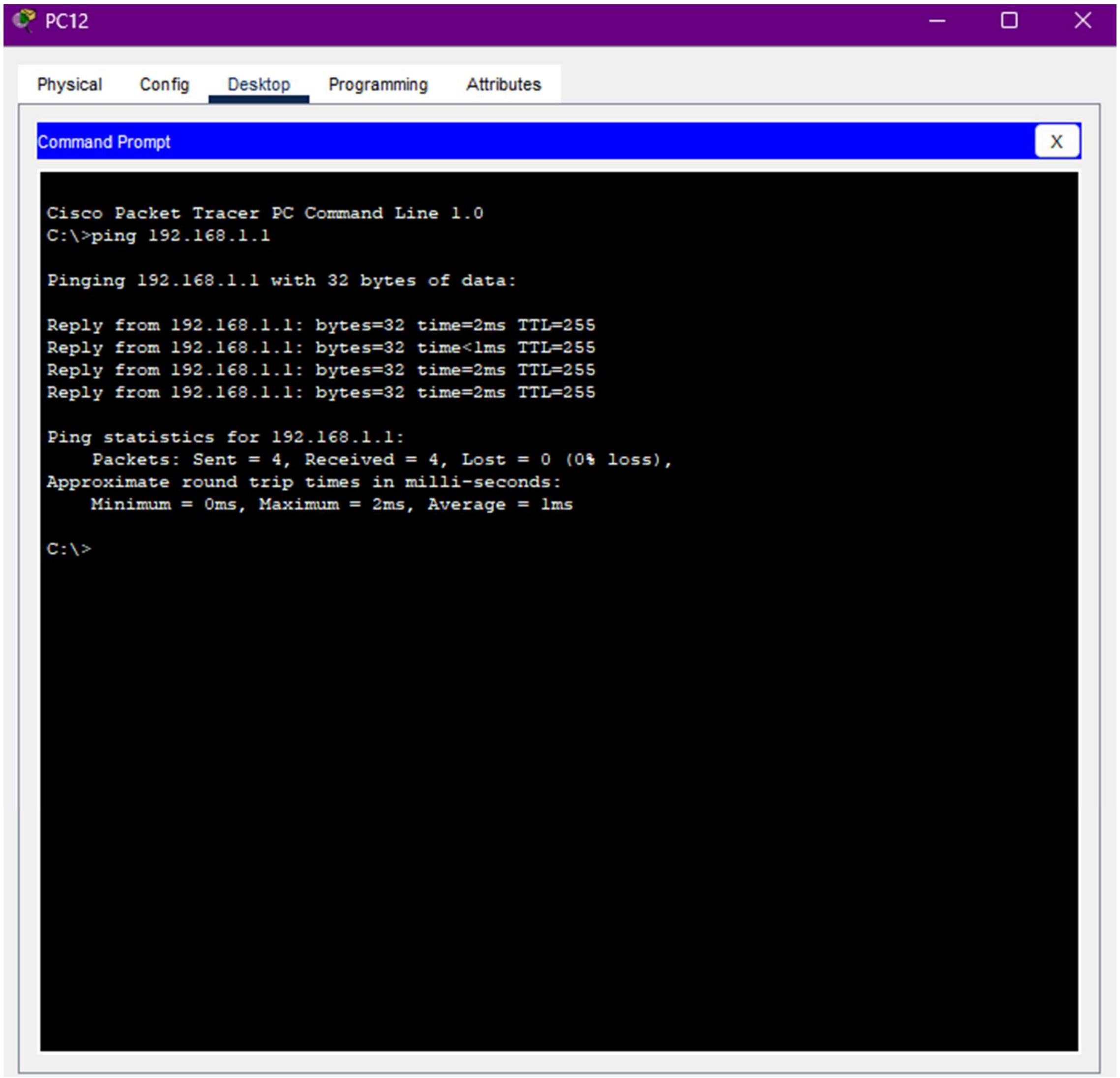
5. NAT Translation


Cloud-PT
Cloud0
2911
Router19
2911
Router18
2960-24TT
Switch9
PC-PT
PC12
PC-PT
PC14
Clo


Router18
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>
Router>
Router>
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#ip address 200.0.0.2 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
Copy
Paste


Router19
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
Router>
Router>
Router>
Router>
Router>enable
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#ip address 200.0.0.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 200.0.0.2
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Copy
Paste




```
IP NAT debugging is on
Router#
Router#
Router#
Router#
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 3  Misses: 4
Expired translations: 4
Dynamic mappings:
Router#
```

```
Router#
Router#
Router#
NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [17]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [4]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [18]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [5]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [19]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [6]

NAT: s=192.168.1.20->203.0.113.2, d=203.0.113.10 [20]

NAT*: s=203.0.113.10, d=203.0.113.2->192.168.1.20 [7]
```